

EXP.-1 INTRODUCTION TO DIGITAL SIGNALLING

EQUIPMENT REQUIRED

U-2970A Data Source
U-2970H Data Receiver
U-2970K Audio Module
U-2970M Power Supply
U-2970N set of connecting leads

Function Generator
Two-channel oscilloscope

OBJECTS:

To show how data represented by data words can be sent as a digit stream, one bit at a time and reconstructed at a distant receiver.

To show that analog signals can be converted to data words and sent by this process.

To show that communication (for example between telephones) can be accomplished in this way.

INTRODUCTION

In order to learn what we mean by a digit stream and many of the words associated with it, we shall build up a digital system, part by part. At each stage we shall study it using some very simple signals.

When it is all working, we shall see that the system can transmit audio signals.

METHOD

1. Establishing the Data Source

Connect the Data Source Module U-2970A into a power supply unit U-2970M and see that the power is on. (This is a general requirement for all modules except L, and will not be mentioned again).

Make the connections and the two switch settings shown in Fig 2-2.

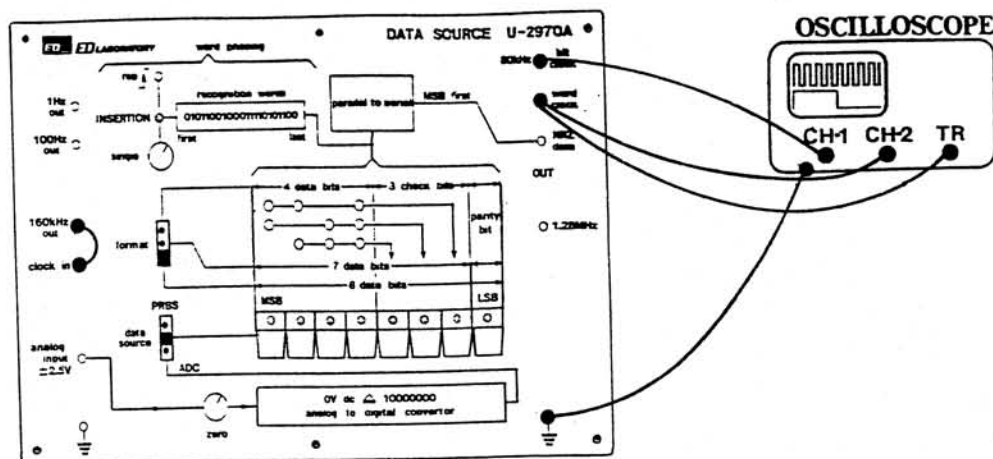


Fig 2-2 Bit and Word Clock Signals

Set the oscilloscope as follows:

CH1 and CH2 d.c.-coupled; 5V/division

Time Base; 10us per division, externally triggered by +ve going edge.

The oscilloscope should produce a display as indicated in Fig 2-2. Adjust it if necessary until it does. The Y1 trace shows the BIT CLOCK which determines the rate at which bit signals will be generated. (We shall see these signals in a moment). The lower trace shows a clock which re-cycles after 8 bit-times. It is called the WORD CLOCK and marks the start of a group of eight bits called a WORD. (People may use other names, discussed in Appendix A).

Move the CH2 connection down to the 'NRZ data' socket, Fig 2-3. The CH2 trace may simply show a straight line at this stage.

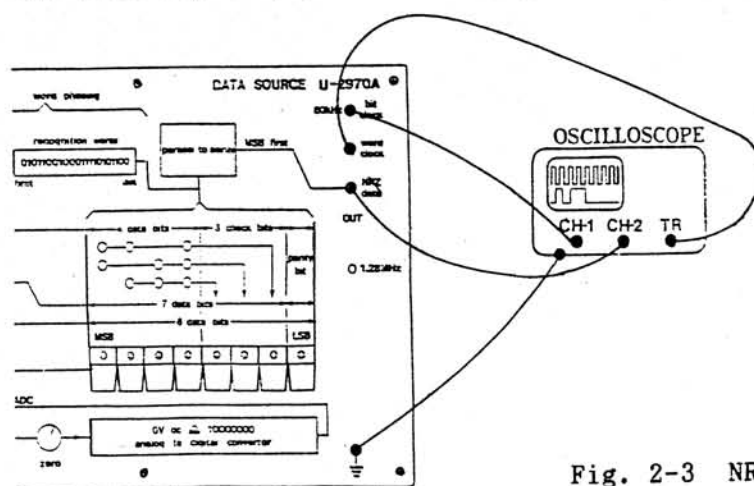


Fig. 2-3 NRZ Data Signal

There is a row of black push-buttons on the module. Each button controls one 'bit' of the signal which will be sent. A bit, or binary digit, is a signal which may be in either of two states, conventionally labelled 0 and 1. Pressing a button will change the state of its bit, indicated by a small lamp being lit for a '1' and dark for a '0'.

Use the buttons to set up the pattern 01001100. Notice that as a button is pressed, the CH2 waveform changes. The CH2 waveform is a simple form of digital signal in which a stream of 'bits' is sent, one after the other. The word set up on the pushbuttons presents all eight bits of the word at once, 'in parallel'. They are applied to a 'parallel to serial converter'. This sends each of the input bits in turn to the 'NRZ data' output, timed by the bit clock.

2. Sending an Analog Signal Digitally

In order to send an analog signal it must first be converted into digital form. The Data Source module has an analog -to-digital converter (ADC) in it. Select the ADC, using the switch near the lower left corner, Fig 2-4.

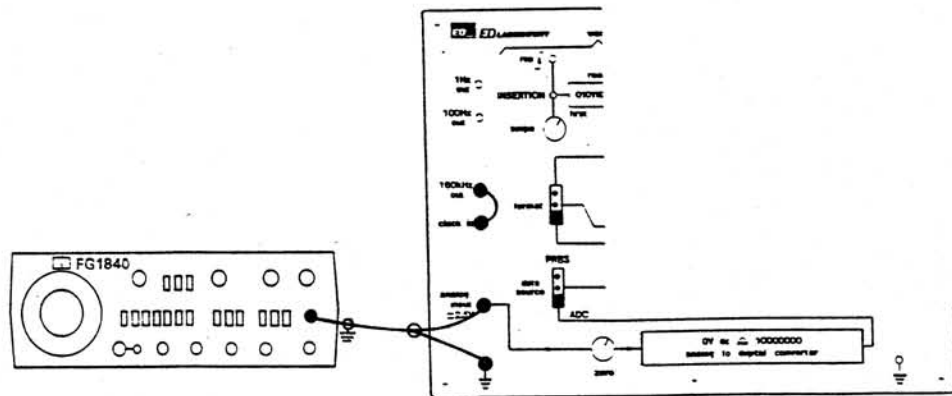


Fig 2-4 Analog Input to Digital System

The bits of the signal are now determined not by the push buttons, but by a voltage at the 'analog input' socket. For the moment this is unconnected. Use the small 'zero' knob to adjust the data word to 10000000.

Next we will connect a function generator (Fig 2-4) to provide an analog signal. Set the function generator to produce:

- Triangle Waveform
- 4V peak-to-peak
- 0.01Hz

Connect the function generator to the 'analog input' and earth (ground) terminals. The behaviour of the data word should now be recognisable as binary counting (reversed at the end of each slope of the triangle input). This is because as the analog signal input changes, the ADC correspondingly generates binary numbers. The NRZ data output varies accordingly. Fig 2-5 shows part of the sequence of binary numbers.

0	1	1	1	0	1	1	1
0	1	1	1	1	0	0	0
0	1	1	1	1	0	0	1
0	1	1	1	1	0	1	0
0	1	1	1	1	0	1	1
0	1	1	1	1	1	0	0
0	1	1	1	1	1	0	1
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	1
1	0	0	0	0	0	1	0
1	0	0	0	0	0	1	1
1	0	0	0	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	0	0	1	1	0
1	0	0	0	0	1	1	1
1	0	0	0	1	0	0	0

Notice how, going down the table, each digit changes half as often as the one to its right.

Fig 2-5 Part of the Sequence of Binary Numbers

3. Receiving Digital Words

Add the Data Receiver module, U-2970H, by placing it to the right of the Data Source connecting the two together, and setting the two switches on the Data Receiver as in Fig 2-6.

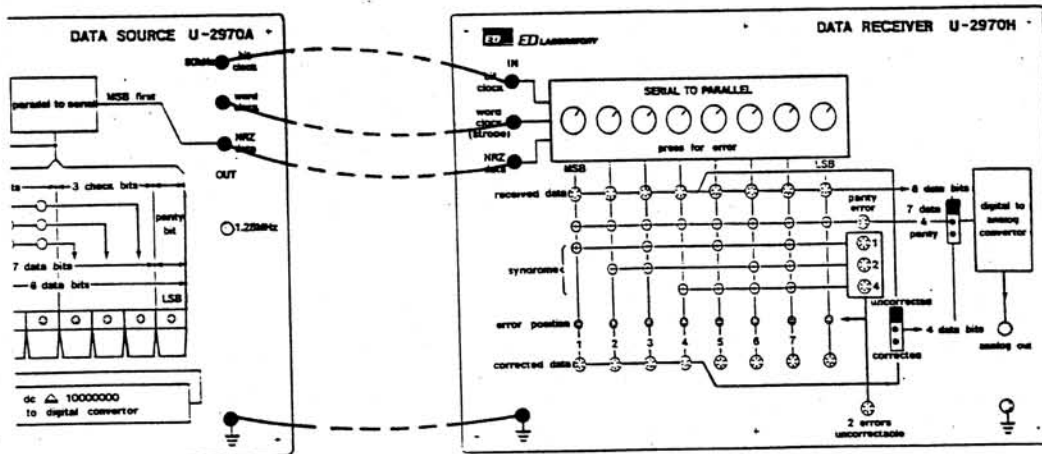


Fig 2-6 Data Receiver

If everything is correct, the 'received data' lamps in the Receiver module should now show the same patterns as are being sent from the Source. This should still apply if you alter the frequency of the input signal. Try raising it (but do not raise the input voltage above 5V peak-to-peak).

Note that the clock signals are important. If you disconnect either of them the Receiver will stop receiving data. The reason for the two clock signals (and ways of doing without two extra wires to connect them) will be the subject of later experiments.

4. Obtaining an Analog Output

If (as is usual) the output to be presented to an analog device, the received word is processed by a digital-to-analog converter (DAC). The DAC on the U-2970H Receiver module is permanently connected.

Move the oscilloscope connections:

- CH1 to analog input of Data Source
- CH2 to analog output of Data Receiver
- trigger to function generator (auxiliary TTL square wave output of FG-1840 is ideal).

The time base will need adjustment to match the frequency of the function generator. If the latter is at 100Hz, the time base may be 20ms per division, for instance.

We now have a simple digital communication system which will accept an analog input signal, convert it to a stream of digits for transmission, send it to a remote location, and produce an analog output which is a reasonably good copy of the original input. We could use it to send a telephone signal for instance.

5. Operating the Audio Module U-2970K

The Audio Module U-2970K can work either as a microphone (receiving sound and sending an electrical signal), or as a speaker (receiving an electrical signal and sending out a sound). Connect the input terminals of an Audio Module to the function generator and set its switch to 'speaker'. (Do not forget the power supply connections). Probably not very much will be heard at first.

Raise the frequency of the function generator to about 600Hz and adjust the 'level' control on the module. A tone should now be clearly audible.

6. Completing a Digital Audio Channel

The function generator was linked to the speaker by a pair of wires whose function was to impress on the 'speaker input' terminals the voltage present at the generator output terminals. But this function could also be done by the digital system made up of the Data Source and Data Receiver modules.

Without disturbing the connections between the Data source and Data Receiver modules, disconnect the direct link to the speaker and substitute the digital data link as shown in Fig 2-7. If this is done correctly, the speaker's behaviour should be much as before.

This shows that an audio signal can be sent through the Data Source and Data Receiver modules. But why bother, you may ask, when a simple pair of wires will do the same? This question will be discussed in Experiment 2.

Try the effect of the digital channel on the sound and the waveform of the received signal as the frequency is increased. It will be found that, as the frequency goes above 5kHz (half the word clock frequency), different frequencies, called ALIAS frequencies are generated. This is not directly because the system is digital, but is due to the fact that the analog signal is sampled at the word clock rate.

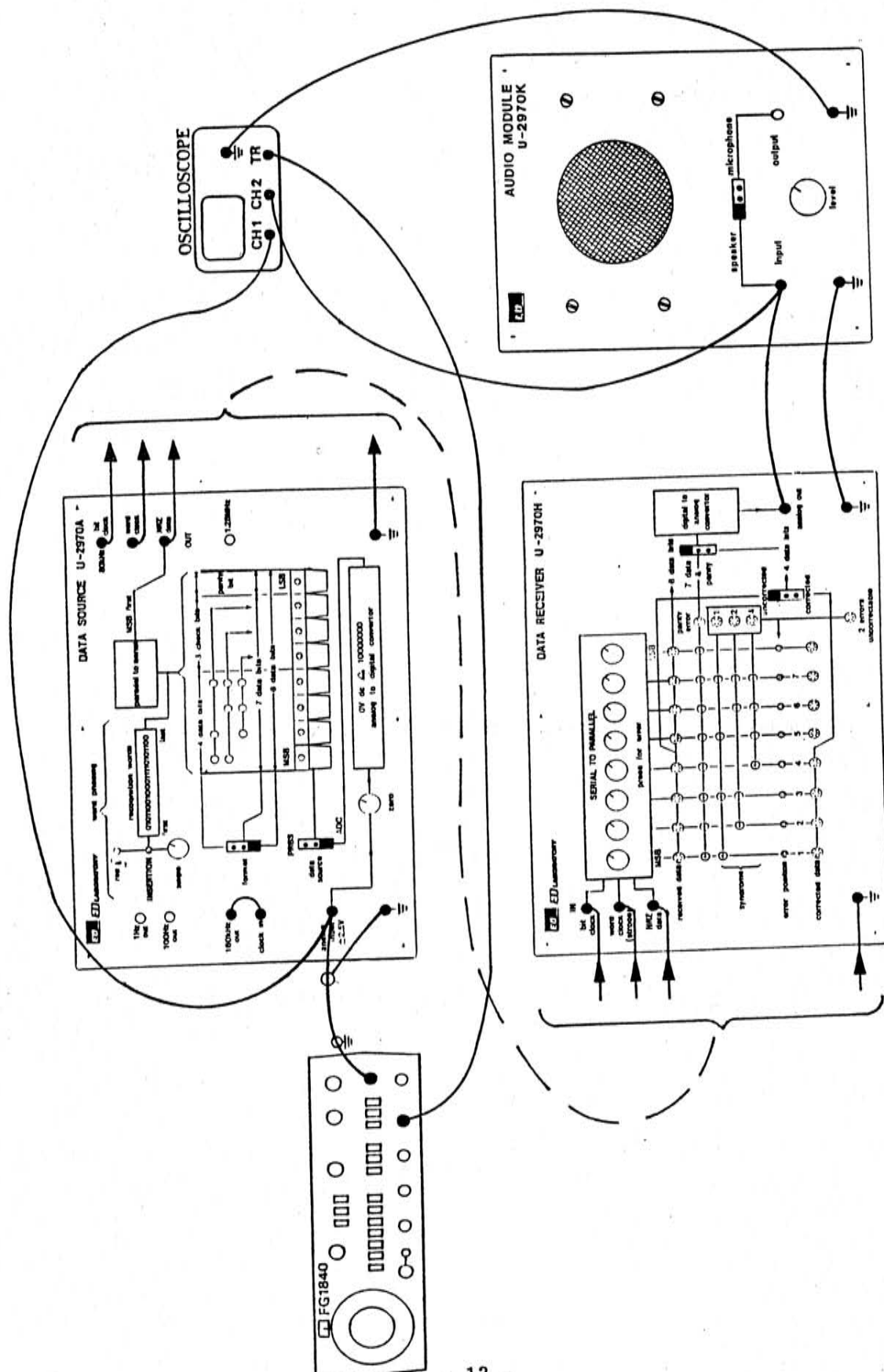


Fig 2-7 Digital Transmission of Audio Signal

7. Testing a simple telephone

If a second Audio Module is available, it can be set up as a microphone and replace the function generator, as in Fig 2-8, thus making a oneway telephone.

Set the Oscilloscope as follows:

- CH1 sensitivity; 0.2V/division
- Time Base; 1ms/division, free-running.

If the level control on the left-hand 'microphone' module is turned clockwise, the oscilloscope should now show a signal in response to any noise entering the module.

If a howling noise results, turn the 'level' control on the right-hand 'speaker' module down (anti-clockwise) until the howl just stops. if you then listen closely to the 'speaker' you should be able hear any sound entering the 'microphone'. The reason for the howling is that sound coming from the speaker is reaching the microphone, which again sends it to speaker so that it goes round a loop repeatedly. if the modules amplify it enough, even the slightest sound gets bigger each time it goes round. To verify that this is nothing to do with the digital system, replace it by a direct wire link between the Audio Modules.

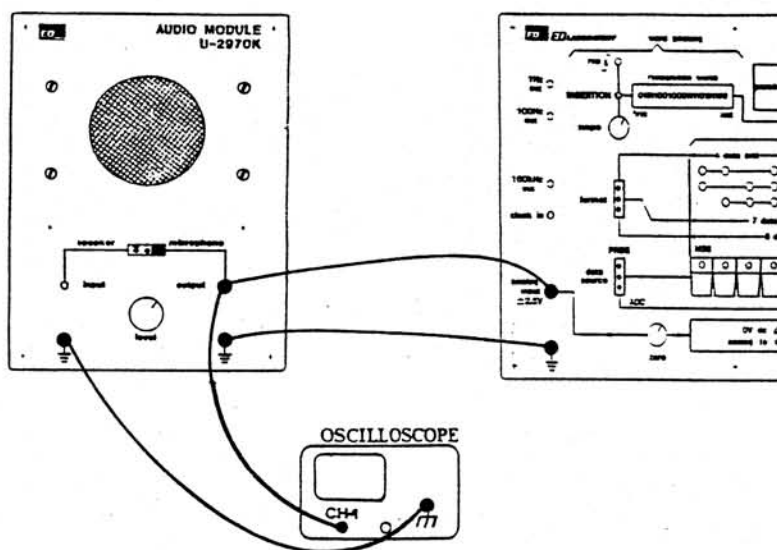


Fig. 2-8 Simple digital Telephone (one-way)